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Effects of Gender and Age on the Levels and Circadian Rhythmicity of Plasma Cortisol*

EVE VAN CAUTER, RACHEL LEPROULT, AND DAVID J. KUPFER

Department of Medicine, University of Chicago (E.V.C., R.L.), Chicago, Illinois 60637 and Western Psychiatric Institute and Clinic, University of Pittsburgh (D.J.K.), Pittsburgh, Pennsylvania 15213

ABSTRACT

Data from rodent studies indicate that cumulative stress exposure may accelerate senescence and offer a theory to explain differences in the rate of aging. Cumulative exposure to glucocorticoids causes hippocampal defects, resulting in an impairment of the ability to terminate glucocorticoid secretion at the end of stress and, therefore, in increased exposure to glucocorticoids which, in turn, further decreases the ability of the hypothalamo-pituitary-adrenal axis to recover from a challenge. However, the consensus emerging from reviews of human studies is that basal corticotropic function is unaffected by aging, suggesting that the negative interaction of stress and aging does not occur in man. In the present study, a total of 177 temporal profiles of plasma cortisol from 90 normal men and 87 women, aged 18–83 yr, were collected from 7 laboratories and reanalyzed. Twelve parameters quantifying mean levels, value and timing of morning maximum and nocturnal nadir, circadian rhythm amplitude, and start and end of quiescent period were calculated for each individual profile. In both men and women, mean cortisol levels increased by 20–50% between 20–80 yr of age. Premenopausal women had slightly lower mean levels than men in the same age range, primarily because of lower morning maxima. The level of the nocturnal nadir increased progressively with aging in both sexes. An age-related elevation in the morning acrophase occurred in women, but not in men. The diurnal rhythmicity of cortisol secretion was preserved in old age, but the relative amplitude was dampened, and the timing of the circadian elevation was advanced. We conclude that there are marked gender-specific effects of aging on the levels and diurnal variation of human adrenocorticotropic activity, consistent with the hypothesis of the "wear and tear" of lifelong exposure to stress. The alterations in circadian amplitude and phase could be involved in the etiology of sleep disorders in the elderly. (*J Clin Endocrinol Metab* **81:** 2468–2473, 1996)

ONE OF THE MAJOR theories to explain individual differences in the aging process is that the rate of senescence, sometimes referred to as the trajectory of aging, is influenced by cumulative exposure to stress, sometimes referred to as the lifetime's allostatic load. This theory is well supported by animal studies that have demonstrated the existence of a cascade of stress and aging effects on the hypothalamo-pituitary-adrenal (HPA) axis, the major neuroendocrine transducer of stress (1, 2). In rodents, cumulative exposure to glucocorticoids causes degenerative changes in the hippocampus, the brain region that normally inhibits glucocorticoid release. These alterations impair the ability to terminate glucocorticoid secretion at the end of stress, resulting in increased exposure to glucocorticoids and, therefore, an ever-decreasing ability of the HPA axis to recover from a challenge (1).

Whether a similar decline in the ability to recover or the resiliency of the HPA axis characterizes human aging and underlies individual differences in patterns of disease and disability among the ever-increasing population of older adults is unclear. Recent reviews analyzing the findings of numerous, but often contradictory, studies on the effects of aging on basal cortisol levels and/or cortisol responses to stimulation tests tentatively concluded that in man, aging does not seem to be associated with changes in the basal activity of the HPA axis or in its ability to respond to challenge (3–5). However, limited evidence, primarily based on studies of dexamethasone suppressibility, suggests that there may be an age-related impairment in glucocorticoid feedback inhibition of HPA activity (3, 4).

The apparent absence of age-related alterations in basal adrenocortical function in human studies could reflect difficulties in detecting such alterations in the face of the large temporal variability in peripheral cortisol levels. Under basal conditions, the 24-h cortisol profile exhibits a morning maximum, declining levels during the late morning and afternoon, a nocturnal period of low concentrations, and an abrupt elevation after the first few hours of sleep (6). Cortisol secretion is intermittent, with pulses recurring at irregular 1- to 2-h intervals, and the overall wave shape of the 24-h profile is produced by modulation of the height of successive pulses. Studies that have examined the 24-h cortisol profile in young and older adults have consistently indicated the persistence of diurnal rhythmicity, but have been conflicting with respect to the presence or absence of specific abnormalities, such as decreased amplitude of diurnal variation (4). These discrepancies could be related to small sample sizes; heterogeneity of subject populations in terms of sex distribution, age range, and health status; and differences in analytical methods to quantify the circadian wave shape.

In the present study, we collected data obtained in 7 different laboratories and reanalyzed a total of 177 temporal profiles of plasma cortisol from 90 normal men and 87

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Address all correspondence and requests for reprints to: Eve Van Cauter, Ph.D., Department of Medicine, MC 1027, University of Chicago, 5841 South Maryland Avenue, Chicago, Illinois 60637.

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women, with ages spanning 7 decades. This analysis demonstrates the existence of marked gender-specific effects of aging on the levels and diurnal variation of human adrenocorticotropic activity, consistent with the findings from rodent studies showing that stress may accelerate aging.

Subjects and Methods

General methodology

After a search of the literature, 9 groups of investigators who had published data on the 24-h profile of cortisol in normal drug-free subjects and depressed patients were identified and contacted by mail. The present report will consider only normal subjects. Seven of the 9 groups were able to contribute the original data included in their previous publications and completed detailed forms on clinical and laboratory data for each subject. The following identifications are used for the contributing laboratories: University of California-Los Angeles (UCLA), Dr. Robert T. Rubin, 36 profiles (7); Western Psychiatric Institute and Clinic, University of Pittsburgh (WPIC), Drs. David J. Kupfer and David B. Jarrett, 32 profiles (8–10); University of California-San Diego (UCSD), Drs. Joseph Mortola and Samuel S. C. Yen, 7 profiles (11); NIMH, Dr. Thomas A. Wehr, 6 profiles (12); Free University of Brussels (ULB), Drs. Paul Linkowski, Anne van Coevorden, Patrick Biston, and Eve Van Cauter, 37 profiles (13–16); Mount Sinai School of Medicine (Sinai), Drs. Uriel Halbreich and Barney Zumoff, 40 profiles (17, 18); and University of Iowa (Iowa) Drs. Bruce Pfohl and Barry Sherman, 19 profiles (17–22).

Data requested for each subject included sex, age, weight, height, occupation, time of catheter insertion on the study day, time of collection of first and last samples, sampling interval, procedure used for blood withdrawal, mealtimes, times of lights off and on, times of sleep onset and morning awakening, and characteristics of cortisol assay. For women, pre- or postmenopausal status and, when relevant, phase of the menstrual cycle at the time of the study were also requested.

Subjects

Raw data on the plasma cortisol profile were obtained for 177 normal subjects studied in 7 laboratories. All subjects were screened for personal history of depression or other psychiatric conditions, including substance abuse. None of the subjects took any drug. Criteria for inclusion in the analysis included availability of age and sex information for each subject and observation of the 24-h profile of plasma cortisol with a blood-sampling interval not exceeding 30 min. These were obtained for 163 subjects. In addition, parameters quantifying nocturnal cortisol variations were calculated for 14 individual profiles obtained at 20-min intervals during an 8- to 10-h overnight period (8).

Data on weight and height were available for 109 and 96 of 177 subjects, respectively. Table 1 summarizes the data on sex, age, and body mass index (BMI) of the subjects. The groups of men and women were well matched for age and BMI. The majority of subjects (*i.e.* 87% of men and 85% of women) were nonobese (*i.e.* BMI <28.0 for men and <27.0 for women). Those who met the criteria for obesity were only modestly overweight (mean ± SD of BMI for obese subjects was 29.1 ± 0.6 kg/m^2 in men and 29.1 ± 1.7 kg/m^2 in women). For all subjects in whom the BMI could not be calculated, the investigators indicated that the subjects were nonobese.

Pre- or postmenopausal status could be obtained in all but 1 of the 91 women. Sixty-seven female subjects (*i.e.* 77%) were premenopausal, and 19 (*i.e.* 23%) were postmenopausal. Information on phase of the menstrual cycle at the time of the study was available for only 24 of the 67 premenopausal women. The study was performed during the follicular phase in 21 of these 24 women.

TABLE 1. Subject data

	Men	Women
No. of subjects	90	87
Age (mean ± SD; yr)	39.4 ± 16.7	39.6 ± 14.5
Age range (yr)	19 –83	18 –75
BMI (kg/m^2)	24.4 ± 2.7(n=54)	23.4 ± 3.2(n=40)

Cortisol assay

In all studies, plasma cortisol levels were measured by a standard RIA. Reported limits of sensitivity of the assay varied from 2.5–10 ng/mL (*i.e.* 14–28 nmol/L), and intraassay coefficients of variation ranged from 5–12%. Reported interassay coefficients of variation varied from less than 10–17%.

Analysis of individual cortisol profiles

The circadian variation in plasma cortisol was quantified using a best-fit curve based on periodogram calculations (23). The acrophase and nadir were defined, respectively, as the times of occurrence of the maximum and minimum of the best-fit curve. The value of the acrophase/nadir was defined as the level attained by the best-fit curve at the acrophase/nadir. The amplitude of the circadian rhythm was defined as 50% of the difference between the value of the acrophase and the value of the nadir and was expressed in concentration units (nanograms per mL; absolute amplitude) and as a percentage of the 24-h mean (relative amplitude).

Additionally, for each individual profile, two parameters characterizing the timing of the early morning elevation were calculated. The timing of the midmean level was defined as the timing of the first sample after the nocturnal nadir when cortisol levels were at or above 50% of the 24-h mean for at least two consecutive samples. The timing of the onset of the circadian rise was defined as the start of the first significant pulse (as identified by an algorithm for pulse detection (24) after the nocturnal nadir. These two markers of circadian timing are largely independent of the absolute levels of nighttime cortisol concentrations.

Finally, the quiescent period of cortisol secretion was defined as starting when cortisol levels were 138 nmol/L or less (*i.e.* 50 ng/mL) for at least 1 h and ending when cortisol levels were more than 138 nmol/L (*i.e.* 50 ng/mL) for at least 1 h.

Statistical analysis

For each parameter characterizing the 24-h cortisol profile, the possibility that one or more values constituted outliers was examined using the criteria of Grubbs (25), and all calculations were repeated after eliminating significant outliers. In total, only 3 of 177 profiles corresponded to significantly outlying values.

Each of the parameters characterizing the 24-h cortisol profile was considered as a dependent variable in analyses of covariance, including sex as factor, and age and BMI as covariates. The laboratory of origin was not used as an additional main factor because, as the age and sex distribution of the subjects varied from one laboratory to the other, effects of this factor could not be interpreted as independent of either sex or age effects. To examine possible interlaboratory differences independently of the effects of age and sex, an ANOVA with laboratory of origin as the independent variable was performed separately for young men (n = 31) and young women (n = 25) under 30 yr of age. This analysis indicated that interlaboratory differences were nonsignificant for all parameters except the timing of the morning acrophase (men, $P < 0.001$; women, $P < 0.005$), which occurred earlier in subjects studied at the WPIC (acrophase at 0601 ± 63 min), and later in those studied at the ULB (acrophase at 0758 ± 56 min; $P < 0.0001$).

Unless otherwise indicated, all group values are expressed as the mean ± SD.

Results

Effects of BMI

In this population of lean or modestly overweight subjects, the BMI was significantly correlated with age in men (r = 0.36; $P < 0.01$), but not in women (r = 0.16; $P > 0.30$). Whether used as a covariate with age or instead of age, effects of BMI were nonsignificant for all parameters quantifying the 24-h cortisol profile.

24-h mean and overall pattern of cortisol levels

A progressive elevation in 24-h mean cortisol level from the second to the eighth decade was observed for both sexes (*upper panels* of Fig. 1; $P < 0.0001$). Thus, basal cortisol levels increased by 20 to 50% between 20–80 yr of age. Sex effects were also significant ($P < 0.02$). Premenopausal women had slightly lower 24-h mean cortisol levels than men in the same age range (176 ± 39 *vs.* 197 ± 48 nmol/L; $P < 0.02$). After menopause, this gender difference was no longer significant. The age by sex interaction was near statistical significance ($P < 0.06$), suggesting that the age-related elevation of the mean cortisol level in aging is more pronounced in women than in men.

The *lower panels* of Fig. 1 compare the mean cortisol profiles in men and women under 30 yr of age and 60 yr of age and over. It is apparent that although overall cortisol levels are clearly higher in old age, the qualitative characteristics of the circadian wave shape are preserved. The short term elevation in the midday hours reflects the well known stimulatory effect of the lunch meal on cortisol secretion (26). The cortisol profiles of young men (n = 31) and young women (n = 25) differed significantly, in that in women, the quiescent period was longer (681 ± 154 *vs.* 542 ± 145 min; $P < 0.003$), the level of the morning acrophase was lower (292 ± 66 *vs.* 372 ± 118 nmol/L; $P < 0.004$), and the absolute amplitude of the diurnal variation was smaller (132 ± 33 *vs.* 171 ± 63 nmol/L; $P < 0.01$) than that in men. As demonstrated below, aging affected specific characteristics of the 24-h pattern of corticotropic activity differently for men and women.

Values of the nocturnal nadir, morning acrophase, and circadian amplitude

The level of the nadir increased progressively with advancing age in both men and women (*upper panels* of Fig. 2; $P < 0.0001$). Typically, the cortisol nadir in a man or woman over 70 yr of age was 3- to 4-fold higher than that in a young adult.

Contrasting with the absence of effect of sex differences on nocturnal corticotropic activity, effects of sex differences were detected for morning levels (*lower panels* of Fig. 2; $P < 0.01$). Premenopausal women had morning cortisol acrophases that were lower (316 ± 69 *vs.* 374 ± 107 nmol/L; $P < 0.0005$) and less variable ($P < 0.01$) than those in men in the same age range. In women, there was an increase in the level of the acrophase from early adulthood to old age ($P < 0.002$). In contrast, in men, the level of the acrophase did not increase with age.

Sex differences in age-related changes in the level of the morning acrophase translated into sex differences in the absolute amplitude of the circadian variation ($P < 0.02$). In premenopausal women, the absolute amplitude was lower and less variable in women than in men in the same age range (averaging 140 ± 32 *vs.* 169 ± 58 nmol/L; $P < 0.001$). After menopause, interindividual variability was much larger in women than in men ($P < 0.01$). In women, but not in men, an elevation of amplitude was observed from early adulthood to old age ($P < 0.05$).

When expressed as a percentage of the 24-h mean level, circadian amplitude decreased as a function of age in both men and women at a similar rate ($P < 0.0001$).

Timing of the circadian variation (Fig. 3)

There were no significant effects of either sex or age on the timing of the morning acrophase (*upper panel* of Fig. 3). All three estimations of the timing of the nocturnal minimum (*i.e.* timing of nadir, timing of midmean level, and timing of onset of circadian rise) were advanced with aging. For the timings

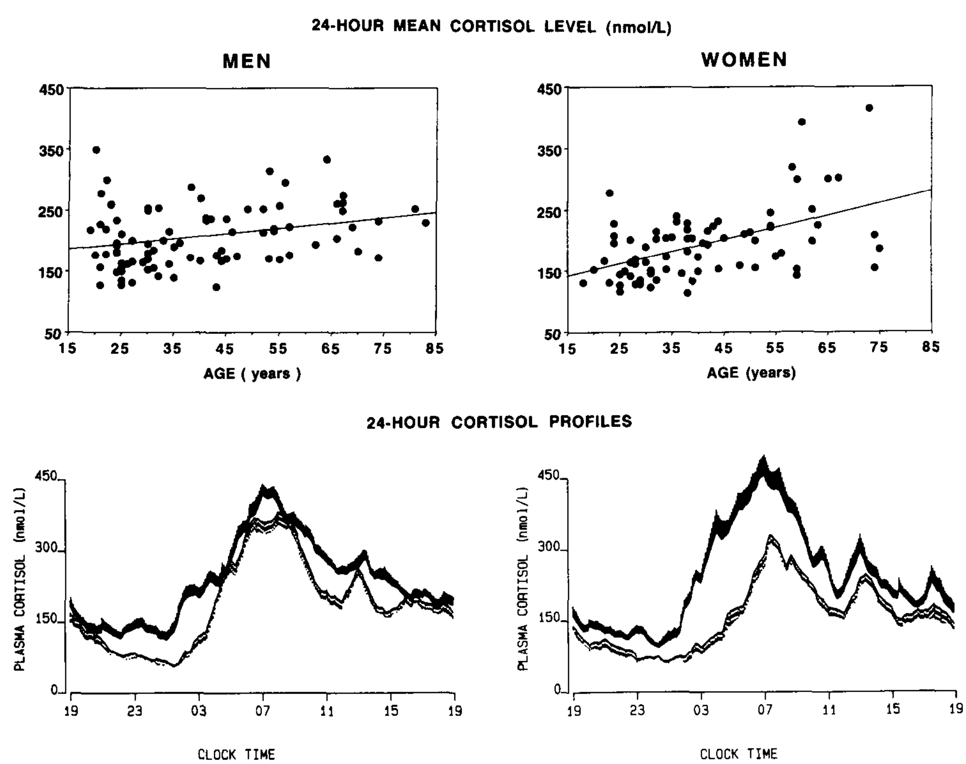

FIG. 1. *Upper panels*, Age-related changes in 24-h mean cortisol levels in men and women. The coefficient of correlation for the linear regression was r = 0.29 in men ($P < 0.002$) and r = 0.50 in women ($P < 0.0001$). Sex differences in the slope of the regression failed to reach significance. *Lower panels*, Mean 24-h cortisol profiles in men and women 50 yr of age and older (n = 25 and n = 22, respectively; *solid lines*) compared to those in 20- to 29-yr-old subjects (n = 29 and n = 20, respectively; *dotted lines*). The *shading* at each time point represents the SEM.

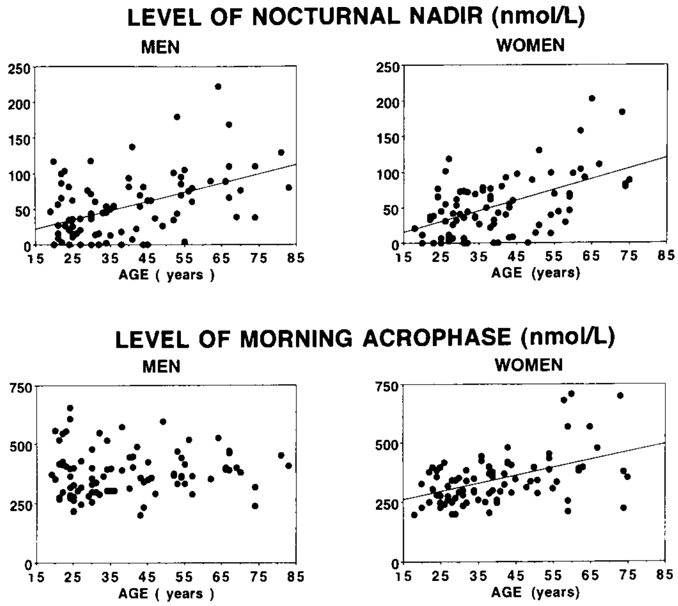

FIG. 2. Age-related changes in levels of the nocturnal nadir (*upper panels*) and levels of the morning acrophase (*lower panels*) in men and women. Similar increases in the level of the nocturnal nadir occurred progressively with aging in both men and women (r = 0.48 and r = 0.53, respectively; $P < 0.0001$). A significant age-associated elevation of the level of the morning acrophase was present in women only (r = 0.48; $P < 0.0001$).

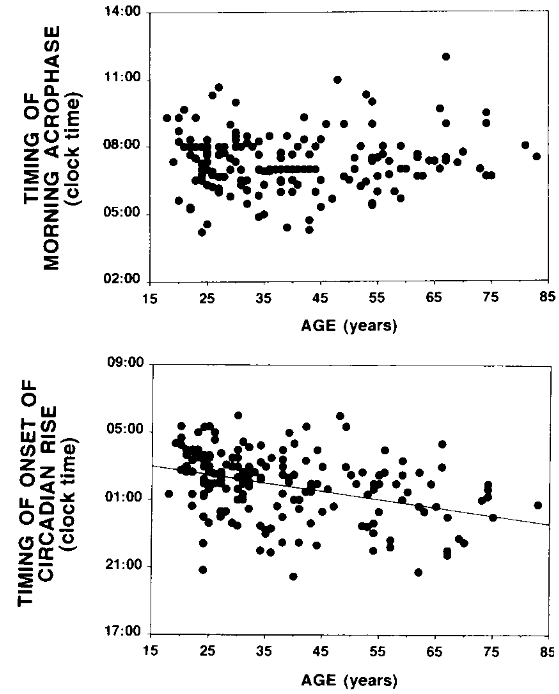

FIG. 3. Age-related changes in estimations of circadian phase. The *upper panel* shows that the timing of the acrophase remained constant across age (r = 0.01). In the *lower panel*, age-related changes in the timing of the onset of the circadian rise (r = −0.33; $P < 0.0001$) are represented, showing a progressively earlier timing of this marker of circadian phase with aging.

of the nocturnal nadir and the midmean level, the coefficients of correlation of the linear regression were, respectively, r = −0.24 ($P < 0.001$) and r = −0.42 ($P < 0.0001$). The *lower panel* of Fig. 3 shows age-related changes in the timing of the onset of the circadian rise. This marker of circadian phase advanced by an average of approximately 3 h between the end of the second decade and the end of the eighth decade ($P < 0.0001$).

Start, end, and total duration of the quiescent period

Throughout adulthood, the quiescent period tended to start earlier in women than in men (*upper panels* of Fig. 4; $P < 0.09$). A progressive delay in the start of the quiescent period occurred with increasing age for both sexes ($P < 0.006$).

Before 30 yr of age, the end of the quiescent period (shown in the *middle panels* of Fig. 4) occurred, on the average, 80 min later in women than in men. This gender difference was no longer significant in later adulthood. Aging was associated with an advance of the end of the quiescent period in both men and women ($P < 0.0001$), but this age effect was more pronounced in women than in men ($P < 0.0002$ for the age by sex interaction). As a result, the quiescent period, which in young adults (<30 yr old) was more than 2 h longer in women than in men, was of similar duration in both sexes after the fifth decade. The reduction in duration of the quiescent period between 25–65 yr of age averaged 161 min in men and 272 min in women ($P < 0.03$). The *lower panels* of Fig. 4 shows the quiescent periods of subjects aged 18–29 yr and subjects 50 yr old and over, illustrating the increased fragmentation, later onset, and earlier offset of the quiescent period in older individuals.

Discussion

The present study demonstrates the existence of marked gender and age effects on basal cortisol secretion and diurnal rhythmicity. Previous studies, involving smaller subject populations, had given contradicting results. However, in a few well documented studies (27, 28), a trend toward an age-related elevation of unstimulated cortisol levels, particularly during the nocturnal period, was apparent, but failed to reach statistical significance due to the small sample size. Therefore, the emerging consensus in careful reviews (3–5) had been that in the absence of stimulus, human corticotropic function under basal conditions was similar in both sexes and unaffected by aging. Based on these negative conclusions, there appeared to be little evidence for an involvement of the HPA axis, the primary neuroendocrine transducer of stress, in mediating deleterious effects of stress on human health and in explaining individual differences in disease susceptibility. Our findings of highly consistent, gender-specific, alterations in the diurnal pattern of basal cortisol secretion across the lifetime strongly suggest otherwise.

Animal studies and a limited number of human studies have indicated that the functional component in HPA regulation that may be most affected by aging is the rate of recovery from a challenge rather than the unstimulated levels, the rate of the initial response, or the magnitude of the response (4, 29). Analysis of the diurnal variation of peripheral cortisol levels under so-called basal conditions in the absence of an identifiable exogenous challenge offers the possibility of examining both the responsiveness of glucocorticoid secretion and the pattern of recovery from stimulation.

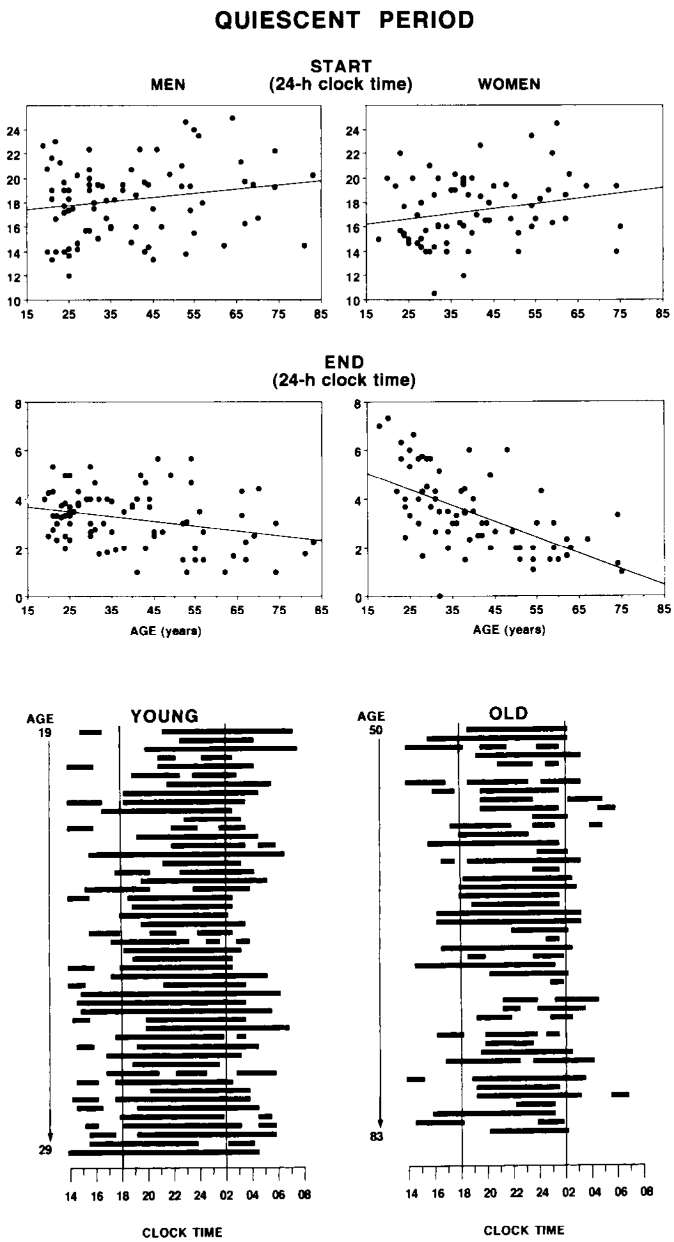

FIG. 4. Effects of aging on the timing of the start (*upper panels*) and the end (*middle panels*) of the quiescent period of cortisol secretion in men and women. With aging, there was a progressive delay of the start (men: r = 0.19; $P < 0.09$; women: r = 0.23; $P < 0.05$) and advance of the end of the quiescent period (men: r = −0.29; $P < 0.01$; women: r = −0.56; $P < 0.0001$). The advance of the offset was more pronounced in women than in men ($P < 0.0002$). The *lower panels* compare the quiescent period (represented by *black bars*) in young (aged 18–29 yr; *left*) and older (50 yr old or more; *right*) subjects. The age ranges are given on the *left* of each graph. *Vertical lines* at 1800 and 0200 h serve to facilitate the visual estimation of the start and the end of the quiescent period. Note the increased fragmentation, later onset, and advanced offset of the quiescent period in older subjects.

Indeed, the sharp early morning increase in plasma cortisol levels represents a response to an endogenous stimulatory signal ultimately originating from the circadian pacemaker located in the suprachiasmatic nuclei of the hypothalamus. The subsequent decline of cortisol concentrations occurring throughout the daytime and evening periods partially reflects the recovery of the HPA axis from this endogenous challenge.

Our analyses indicate that in young adulthood, overall plasma cortisol levels are lower in women than in men because the female response to the circadian signal is slower and of lesser magnitude, and the return to quiescence is more rapid. The possibility that this greater resiliency of the HPA axis in premenopausal women might be involved in their lower rate of cardiovascular and other stress-related diseases deserves further investigation. In men, the profiles of cortisol secretion were characterized by shorter quiescent periods and higher and more prolonged early morning elevations. Interestingly, interindividual variability in the magnitude of the morning elevation was larger in men than in women, suggesting that individual differences in response to stressors may also be more marked in men than in premenopausal women.

During aging, there appeared to be a progressive decline in the endogenous inhibition of nocturnal cortisol secretion in both men and women, as reflected by a delay in the onset of the quiescent period and higher nocturnal cortisol levels. This loss of resiliency is consistent with the hypothesis of "wear and tear" of lifelong exposure to stress and, based on rodent studies, is likely to reflect neuronal loss in the hippocampal area. Such hippocampal defects may underlie some of the memory deficits that occur in a majority of older adults (30). In addition to being relevant to memory impairment in aging, this loss of resiliency could be partially responsible for the sleep disorders of the elderly, as several studies have demonstrated that elevated corticosteroid levels result in increased sleep fragmentation (31, 32).

The data also demonstrate an increased responsiveness of corticotropic activity in older women compared to men in the same age range, consistent with previous observations on the responses to iv CRH administration (33) and to a driving simulation challenge (29). These findings also corroborate rodent data showing more profound age-related changes in HPA reactivity in females than in males (34). Our study further revealed a gender difference in the effects of aging on the recovery rate from circadian stimulation, which was more rapid in women than in men.

The 24-h cortisol profile is a widely used marker of the human circadian clock because of its large amplitude and remarkable reproducibility. The present study demonstrates a progressive advance of circadian phase during senescence, which appears to affect men and women similarly. Our estimations of the relative amplitude of the cortisol rhythm also suggest that the strength of the circadian signal decreases progressively with advancing age in both sexes. These alterations of circadian phase and amplitude could underlie alterations in the sleep-wake cycle of older adults, who tend to have earlier sleep onset, earlier morning awakening, and a more fragmented and more shallow sleep period (35, 36). Studies comparing groups of young and older adults have shown that besides the sleep-wake cycle and the diurnal cortisol profile, a number of other 24-h rhythms, such as those of body temperature (37–39) and plasma melatonin (15, 40), are dampened and/or advanced in old age. Studies performed in sleep-deprived subjects under constant behavioral and environmental conditions have suggested that these alterations result from age-related changes in the output of the circadian pacemaker and are not an immediate response to lifestyle modifications (39).

In conclusion, the present study demonstrates that normal aging is associated with significant alterations in HPA activity, consistent with the "wear and tear" of lifelong exposure to stress. Recent research has indicated that neuronal loss in the hippocampal area is likely to be the primary cause of these alterations and may underlie some of the cognitive deficits that occur in a majority of older adults (30, 41).

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